

DBscreamZ

Pitch-tracking formant scream synthesizer

User manual
& preset field guide

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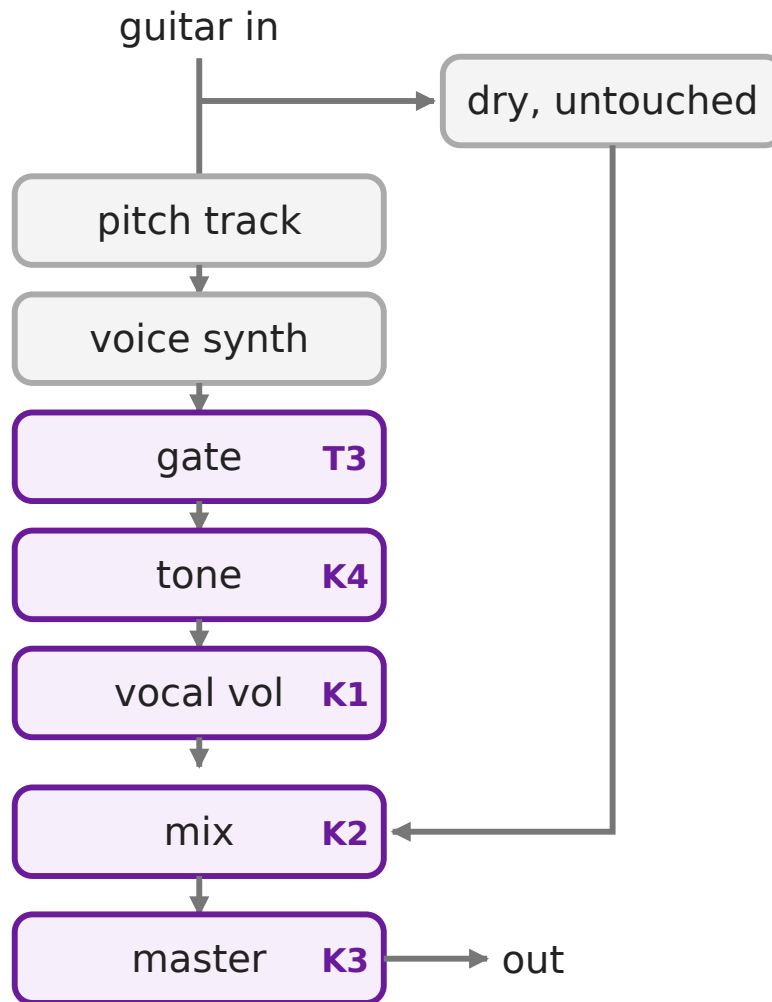
1 What this is

Play a note. A synthesized voice screams it back at you, tracking your pitch as you bend, slide and vibrato.

The voice is not a sample and not a filter draped over your guitar. It is built from scratch as you play: a pitch tracker follows your fundamental, and a bank of formant grains rebuilds a vowel at that pitch. Three formants, F1, F2 and F3 [formant 1, 2 and 3], decide which vowel and how bright it is.

Your dry signal is never processed. It is passed through untouched and crossfaded against the voice at the end, so Mix at 7:00 with Master at 12:00 is your guitar, exactly as it went in. Master sets the level of both.

Signal flow



The gate watches your input envelope and decides when the voice may sound. It never gates your dry signal.

K1 to K6 [knob 1 to knob 6] are the knobs and T1 to T3 [toggle 1 to toggle 3] the toggles, numbered left to right.

2 First sound

1. Guitar into **IN**, amp into **OUT**, 9 V DC [direct current] into the barrel jack. The pedal boots **bypassed**, both LEDs off.
2. **T2** (the middle toggle) **up** (Set 1). Put **T1** and **T3** in their **middle** position.
3. Tap the **RIGHT** footswitch. The right LED goes solid: Wukong is engaged.
4. Play single notes, cleanly, one at a time. The voice follows your pitch.
5. Nothing screaming? Turn **knob 2 (Mix)** clockwise and set **T3** (the gate) to its **middle** position.
6. Now stomp **both** footswitches at once and hold. That is Charge mode.

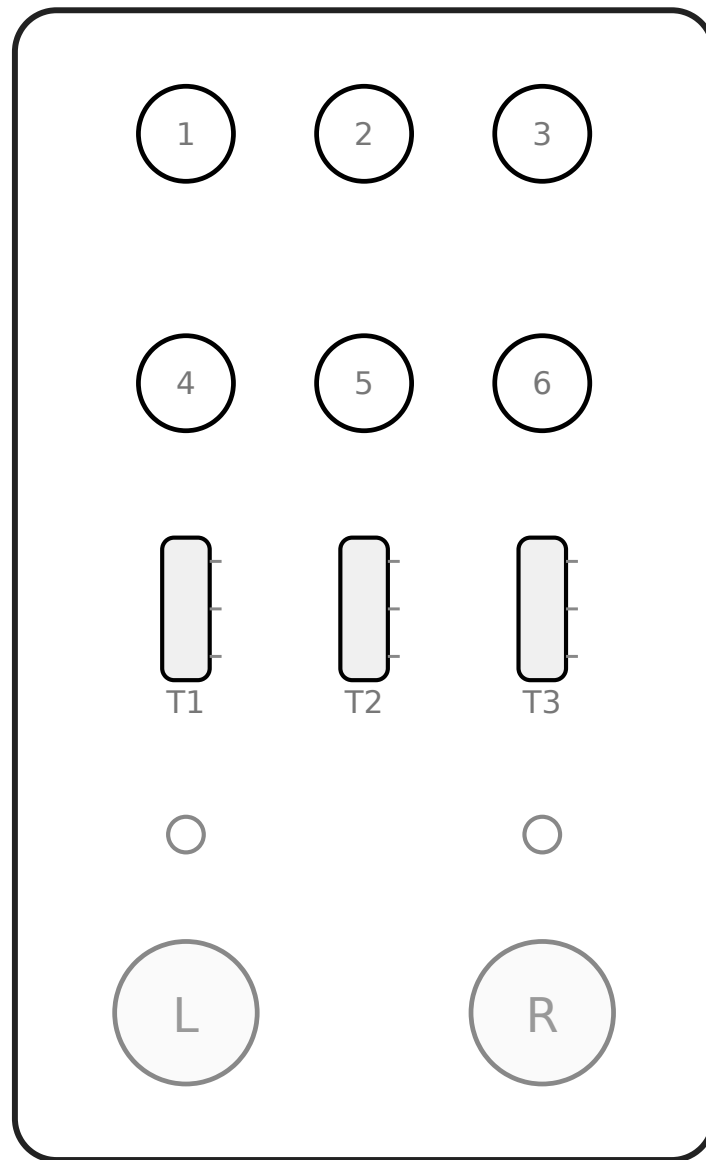
ONE NOTE AT A TIME

Pitch tracking needs a single fundamental. Chords, open strings ringing under a lead line, and heavy pick noise all confuse it. Mute what you are not playing.

Tap the same footswitch again to bypass. Tap the **LEFT** footswitch instead for the other voice on this page (Prince). Move T2 to its **middle** position for Set 2 (Rice right, Flute left) or **down** for Set 3 (Master right, Ki-Ki left).

Edits are live at once, and lost at power off or when you tap a voice in, until you save (section 6).

3 The controls



Six knobs in two rows of three, numbered left to right: **1 2 3** on top, **4 5 6** below. Three toggles, each up, middle or down. Two footswitches, an LED [light-emitting diode] above each.

Knob travel

Knobs run **7:00** fully counterclockwise, through **12:00** at center, to **5:00** fully clockwise. Every position in this booklet is a clock face.

What the knobs do

	Menu 1	Menu 2	Menu 3
K1	Vocal vol	F1	F3
K2	Mix	F1 bandwidth	F3 bandwidth
K3	Master vol	F1 amount	F3 amount
K4	Tone	F2	Vib
K5	Glide	F2 bandwidth	Depth
K6	Vocal size	F2 amount	Detune

Menu 3: **K4** (vibrato rate) is cube-tapered over 0 Hz (7:00) to 50 Hz (5:00), so 6.25 Hz sits at the center (**12:00**) and the musical singer's-vibrato range does not get crammed into the bottom of the travel. Every character ships with vibrato off, both **K4** and **K5** fully counterclockwise.

What the menu 1 knobs do

Vocal vol (K1) is how loud the synthesized voice is on its own. **Mix** (K2) crossfades it against your untouched dry signal: 7:00 is the guitar alone, 5:00 is the voice alone. **Master** (K3) is the level of the pair leaving the pedal. Vocal vol and Master are unity (12:00); Mix at 12:00 is an even blend.

Tone (K4) tilts the *voice* dark below 12:00 and bright above it, and never touches the dry. Anywhere from about 11:30 to 12:30 it is exactly flat.

Glide (K5) is how slowly the voice slides to each new note, 0 ms (7:00) to 300 ms (5:00). It is a slide that eases in: at 300 ms (5:00) the voice is most of the way there after 300 ms and settles within about a second. The taper is steep: the first two thirds of the travel stays under 90 ms (1:42), so the whole usable range of slurs sits below 2:00.

Vocal size (K6) scales all three formants together, which is acoustically vocal tract length: the same character shouting the same vowel out of a physically bigger body. At **7:00** it is the character exactly as written, which is what every card in section 11 assumes. At **5:00** every formant is halved, the deepest it goes.

What the toggles do

	Up	Middle	Down
T1 Octave	+1	0	-1
T2 Page	Set 1	Set 2	Set 3
T3 Gate	high	medium	low

That is with no menu latched. While a menu *is* latched the toggles configure Charge mode instead, and nothing else. See section 7.

A toggle acts when you *move* it. A voice you tap in brings its own saved octave and gate, whatever the switches say, until you move one.

The gate decides how loud you must play before the voice speaks. High needs a firm attack and stays quiet between notes; low lets quiet playing through and hangs on longer.

4 Every gesture

To do this	Do that
Engage	Tap either footswitch
Bypass	Tap the lit side again
Switch voice	Tap the other side
Latch menu 2 (F1, F2)	Hold RIGHT ~1 s: it latches while you hold
Latch menu 3 (F3, vibrato, detune)	Hold LEFT ~1 s: it latches while you hold
Jump to the other menu	With one latched, tap the other footswitch
Leave a menu	Tap the blinking side's own footswitch
Save to a slot	Hold one footswitch past 1 s, then press the other while still holding. The held side is the slot.
Charge	Engaged, no menu: stomp both together and hold
Firmware update	Bypassed only: press both together, hold ~2 s

TOGETHER, NOT STAGGERED

Charge needs both switches going down at the same time. Holding one for a second and then adding the other is the *save* gesture. Bypassed, any 2 s with both down is update mode, and a stagger of a second or more also saves the held side.

What the LEDs mean

Single color, both of them: the left LED is always blue, the right always orange. Nothing ever changes an LED's color, only whether it is off, solid or blinking.

State	Left (blue)	Right (orange)
Bypassed	off	off
Left slot engaged	solid	off
Right slot engaged	off	solid
Menu 2 latched	off	blinking
Menu 3 latched	blinking	off
Saved	3 blinks	3 blinks
Charging	alternating, speeding up as it builds, slowing as it winds down	

5 The three menus

Six knobs, eighteen parameters. What a knob does depends on which menu is latched.

Menu	How to get there
1	Where you are with nothing latched
2	Hold the RIGHT footswitch. About 1 second in it latches, while you are still holding: right LED blinks, left goes dark. Letting go changes nothing.
3	Hold the LEFT footswitch the same way. Left LED blinks, right goes dark.

A menu **latches**: it stays until you leave it.

- Tap the **other** footswitch to jump straight to the other menu.
- Tap the **blinking side's own** footswitch to drop back to menu 1.

KNOB PICKUP: NOTHING JUMPS

Every time you change menu, or recall a voice, all six knobs go **inert**. A knob does nothing until you **move it**, about a fingernail's nudge. The moment it moves it takes over its parameter, from wherever it is physically sitting.

A knob you do not touch cannot hurt the sound. A knob you do touch grabs control at once: it does not wait for you to sweep back to the stored value.

What the formant controls mean

F1 and F2 decide which vowel the voice is shouting. Move them and the scream changes from an *ah* to an *ee* to an *oh*.

F3 mostly decides how bright and nasal it sounds.

Bandwidth is how wide each resonance is: narrow is more vocal, wide is more washed out. **Amount** is how much of that formant you get.

Vibrato rate and **Vibrato depth** add a periodic wobble on top of the pitch you are already playing: rate is how fast it cycles, 0 Hz (7:00) to 50 Hz (5:00), and depth is how far it swings, 0 (7:00) to 100 cents (5:00). Every character ships with both off (7:00).

Detune spreads the engine's three voices apart: the outer two move up to 60 cents (5:00) either side of the middle one. Wider detune thickens and roughens the scream; at 0 cents (7:00) the three collapse to the bare, focused character.

6 Memory

T2 (the middle toggle) picks a **page**. Each page has two **slots**, one per footswitch. Six stored voices.

T2	Page	Left	Right
Up	Set 1	Prince	Wukong
Mid	Set 2	Flute	Rice
Down	Set 3	Ki-Ki	Master

Those six are what it ships with. Save over any of them: the recipes to dial them back are in section 11.

Saving

Everything you tweak is live but volatile. To save:

1. Hold **one** footswitch past 1 second.
2. **While still holding it**, press the other.

The **held** side picks the slot. Hold RIGHT and press LEFT and you have saved to the **right** slot of the current page. Both LEDs blink three times.

Leave any latched menu before saving. Menus never save, and your edits survive the exit.

The held footswitch latches its own menu on the way past 1 second, the same as any hold. The second press drops that menu again and saves, so you finish where you began.

7 Charge mode

With a voice engaged and **no menu latched**, stomp **both** footswitches together and hold. The scream charges: gain swells, pitch sweeps, the voice grows, and the LEDs alternate faster and faster as it builds. Release and it winds down.

At a **full charge every setting you have switched on goes as far as its position takes it**. For gain and size, up goes to the top of the range (gain all the way up, the voice at its deepest) and middle goes halfway from wherever the voice already sits to that top. Tone is a direction rather than an amount, so both of its positions go the whole way: darker is fully dark, brighter fully bright.

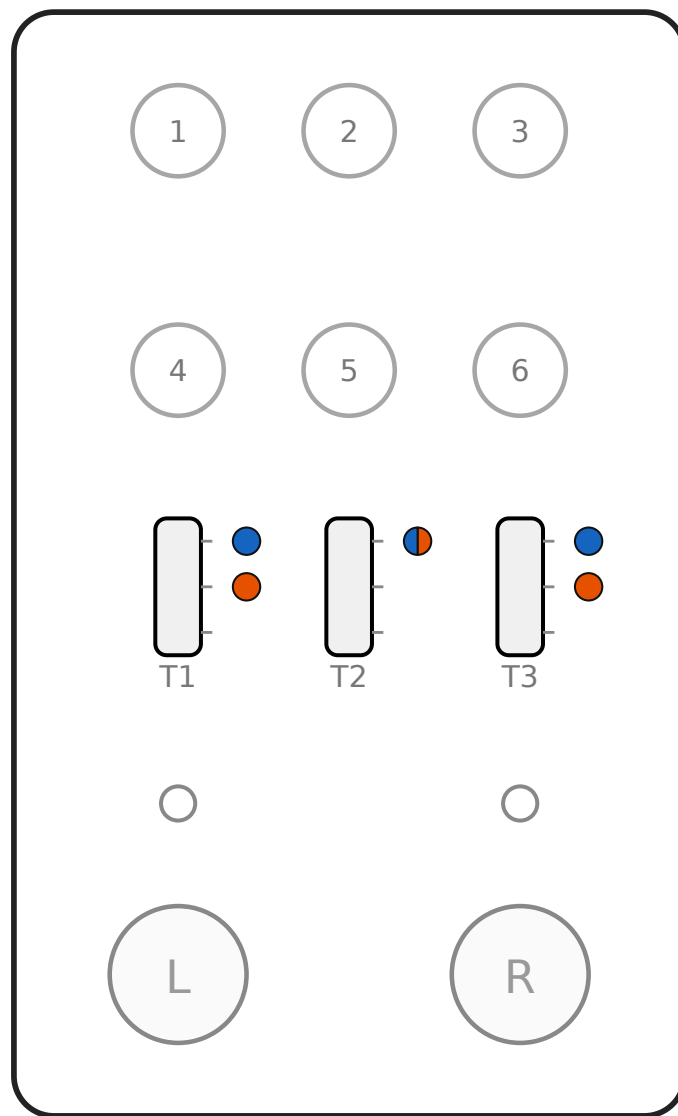
Pitch glides toward two octaves away for as long as you hold: nearly there by the end of the slowest charge (Birit Spomb, ~6 s), about one octave in by the end of the quickest (punch, ~0.75 s). The two octaves count from wherever the octave switch sits, so a voice whose switch is already up rises three octaves above normal.

It lasts exactly as long as you hold it. No latch, no timeout.

Engaged is the whole condition. The same two-stomp hold while *bypassed* is the firmware-update gesture instead, so charge can never reach it and it can never reach charge. See section 12.

Charge is configured on the **toggles**, while a menu is latched. Latch menu 2 (hold RIGHT) for gain, charge time and decay, menu 3 (hold LEFT) for pitch, tone and size. The knobs are not part of charge. The setting is stored when you leave the menu by tapping the blinking side; power off with a menu still latched and the change is lost.

Factory charge settings



Out of the box it is the full power-up. Orange (menu 2): gain on, Birit Spomb, slow decay. Blue (menu 3): pitch rising, brighter, full size. Each dot sits level with the position it wants; a half blue, half orange dot is both menus wanting the same one. The tables overleaf show the same in bold.

This configuration is **global**: one setting shared by every voice, never touched by saving or recalling. That is why it is not on the character cards.

● With menu 2 latched

	Up	Middle	Down
T1 Gain	Above 9000!	on	off
T2 Charge time	Birit Spomb ~6 s	Hamekameka ~2.5 s	punch ~0.75 s
T3 Decay	fast	slow	off (instant)

● With menu 3 latched

	Up	Middle	Down
T1 Pitch	rise 2 octaves	fall 2 octaves	off
T2 Tone	brighter	darker	off
T3 Size	full	half	off

Factory positions are marked in bold. A toggle sets charge when you *move* it, so to pick the position it already sits in, flick it away and back. For a charge that *falls* instead of rising, latch menu 3 and center T1; for a quicker build, latch menu 2 and center or drop T2. Your Set, octave and gate stay as they were; the switches no longer show them until you next move them with no menu latched.

8 Chord mode

Hold both footswitches through power-up and the pedal spends that session as a single vowel filter driven straight off your guitar, chords included, instead of the six characters. There is no pitch tracker in this path: it cannot make an octave error, and at octave 0 it adds no delay. Toggle 1 shifts everything an octave down or up, and charge sweeps it two more.

Entering

Hold both footswitches down and keep holding as you power the pedal on. Both LEDs flash three times to say chord mode is live. Power off and back on **without** holding the footswitches and the six characters are exactly as you left them.

Stomp	Does
LEFT tap	Engage or bypass. In the chord menu, leaves it instead.
LEFT hold ~1 s	Latch the chord menu. Left LED blinks. Tap LEFT again to leave.
RIGHT, held	Open the mouth: right LED lit for as long as you hold it (dark in the chord menu). Momentary, never a menu, never a save.
Both together	Charge, engaged and outside the menu only: the same gain, pitch, tone and size overlay, set up in normal mode. Pitch sweeps two octaves from wherever toggle 1 sits.

Toggles and saving

Toggle 1 is the octave: up +1, middle 0, down -1, saved with the chord setting like the characters' octave. Toggle 3 is the gate, same high/medium/low as the characters. Toggle 2 does nothing in chord mode. While shifted there is a slight delay (about 20 ms) and a little grain, more the further you go; at octave 0 there is none.

Chord mode has one setting, not six slots, and it saves itself: a few seconds after you stop turning a knob, or at once when you leave the chord menu, and only when something actually changed. There is nothing to press. It lives in its own corner of memory; your saved characters are never touched by it.

The two knob layers

	Main (default)	Chord menu (hold LEFT)
K1	Vocal vol	Closed vowel
K2	Mix	Open vowel
K3	Master vol	Vocal size
K4	Tone	Resonance
K5	Sensitivity	Attack
K6	Drive	Release

The vowel knobs each pick one of five vowels, oo, oh, ah, eh, ee. Play softly and you hear the closed vowel; the harder you pick, the further it moves toward the open vowel. Sensitivity (K5) sets how hard you must pick to reach it. Hold RIGHT and it goes straight to the open vowel.

Main layer, every position

	Knob	Range, 7:00 to 5:00	Factory	Position	
K1	Vocal vol	0 to 2x	2.0x	100%	5:00
K2	Mix	dry to full voice	full voice	100%	5:00
K3	Master vol	0 to 2x	1.2x	60%	1:00
K4	Tone	dark to bright, flat at center (12:00)	flat	50%	12:00
K5	Sensitivity	0, fixed closed vowel, to 8	1.6	58%	12:51
K6	Drive	1x to 40x, log taper	40x	100%	5:00

Charge and the mouth

While holding RIGHT (mouth open), press LEFT as well to start a charge. The mouth closes while both are down. Let go of LEFT and the charge winds down, and the mouth opens again if RIGHT is still held.

In the chord menu, a LEFT tap with RIGHT held does not count, so let go of RIGHT before tapping LEFT to leave the menu.

Chord menu layer, every position

	Knob	Range, 7:00 to 5:00	Factory	Position	
K1	Closed vowel	oo, oh (9:30), ah (12:00), eh (2:30), ee	oo	0%	7:00
K2	Open vowel	oo, oh (9:30), ah (12:00), eh (2:30), ee	ee	100%	5:00
K3	Vocal size	formants at 100% to 50%, an octave down	100% formants	0%	7:00
K4	Resonance	soft to sharp	0.53	53%	12:18
K5	Attack	1 ms to 50 ms mouth attack	17 ms	69%	1:53
K6	Release	20 ms to 500 ms mouth release	20 ms	0%	7:00

Factory gate: medium. Both layers and the gate are the pedal's one factory-chord setting: the values it ships with out of the box.

The LEDs: both flash three times on entry; left lit is on; right lit is the mouth open; left blinking is the chord menu; alternating is a charge.

9 When it misbehaves

No scream, just my guitar

Check an LED is solid. Then **knob 2 (Mix)**: at 7:00 you hear only dry signal. Then the **gate** (T3): on *high* with low-output pickups the voice may never open. Set T3 to its middle position.

A knob does nothing

Working as designed. After every menu change and every recall the knobs are inert until moved. Turn the knob a little further.

It chases the wrong note

Play one note at a time and mute what you are not using. Open strings ringing under a lead line are the usual culprit. Heavy drive in front makes tracking worse: put DBscreamZ before the drive in your chain.

I tried to save and got update mode

While bypassed, both footswitches down for 2 s starts update mode, even if one went down first. The save happens the moment the second goes down, so release both at once, or engage a voice first: update mode cannot start while one is engaged.

It sounds thin and buzzy

Formant bandwidths set too wide will do that. The cards give the factory values: F1 32.5 Hz (7:56), F2 47.5 Hz (8:05) and F3 62.5 Hz (8:10).

The scream lags my playing

Check **glide** (knob 5). At 5:00 the voice takes most of a second to settle on each new note. At 7:00 it is instant.



The rest of this book is the voices. Each of the eight characters is a card with every knob in all three menus and every toggle. Six come saved in the pedal, two to a Set; the other two are yours to dial in, then two blank cards for voices of your own.

10 Reading a card

Each card is the pedal face with a pointer drawn out of every knob, one per menu, ending at the position that menu wants. **Twenty-one positions:** six knobs in each of three menus, plus three toggles. Set every one. None can be assumed: the knobs are wherever you left them.

-  **Menu 1**, solid pointer
default layer
-  **Menu 2**, dashed pointer
hold RIGHT
-  **Menu 3**, dotted pointer
hold LEFT

Told apart by color *and* by line, so a photocopy still works. Menus 2 and 3 wear the color of the LED on the side you hold to reach them: orange for the right, blue for the left.

Toggles on a card

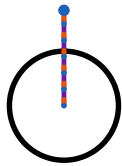
Toggle dots sit to the right of the switch, level with the position they want.

Every character wants octave 0 and the medium gate: the middle position on T1 and T3.

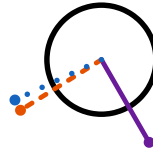
The T2 dot and the footswitch drawn with a heavier outline show the factory home: its Set and its slot. You can save a voice anywhere.

Two of the eight have no home and no T2 dot: dial them in and save them where you like. After them are two blank cards for voices of your own.

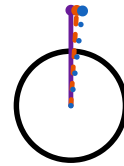
Every pointer sits at its exact position, including when menus agree. The line styles interleave, so where two or three menus want the same knob in the same place you get one pointer striped in their colors rather than three lines fighting over the same pixels.



All three agree. Knob 3 on every card: 12:00 in all three menus.



One apart, two close. Knob 2 on every card.



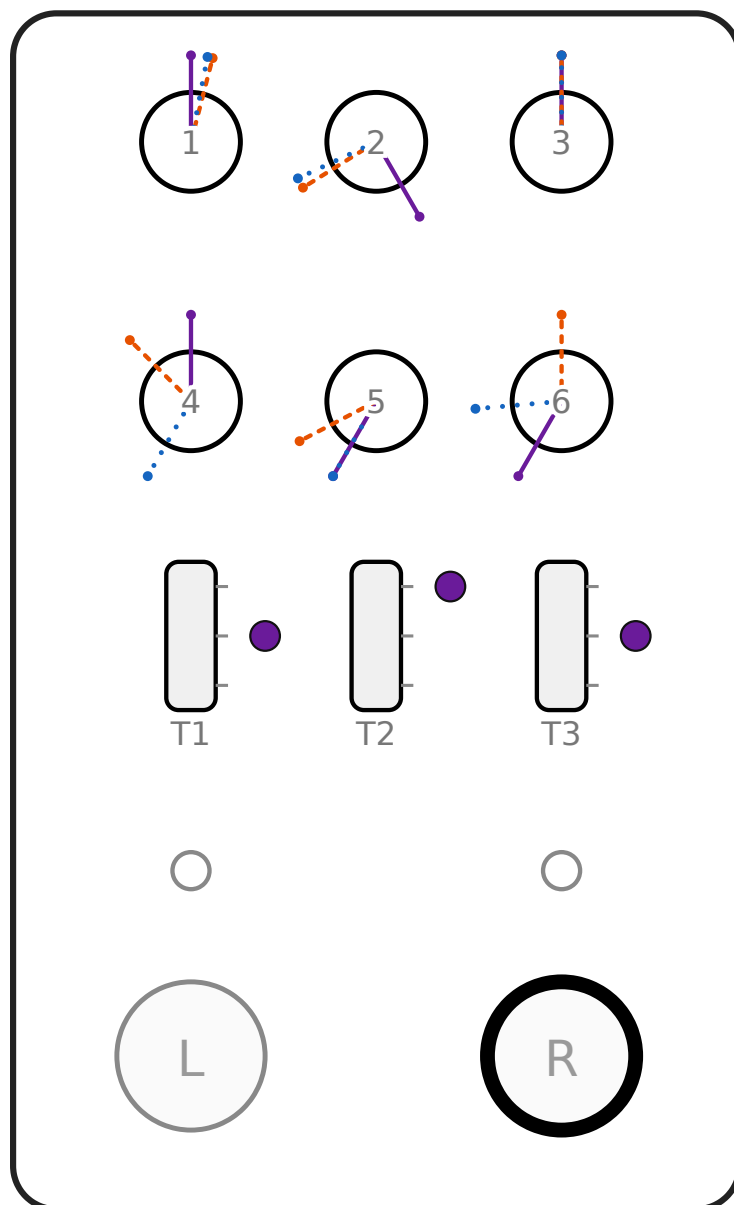
A tight cluster. A worked example, not any one card's knob: three positions inside 14 minutes.

THE TABLE IS THE EXACT WORD

Pointers are drawn true, but a knob is a blunt instrument and two pointers a few minutes apart look like one. Each card is followed by the same twenty-one positions as a table. When the drawing is ambiguous, read the table.

Wukong *The default shout*

Factory: Set 1, RIGHT slot



Mid-placed formants with a narrow detune. The straightest read of the effect: loud, open, clean. Start here.

Wukong: every position

● Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

● Menu 2 (hold RIGHT)

K1	F1	858.4 Hz	55%	12:29
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1234 Hz	35%	10:30
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

● Menu 3 (hold LEFT)

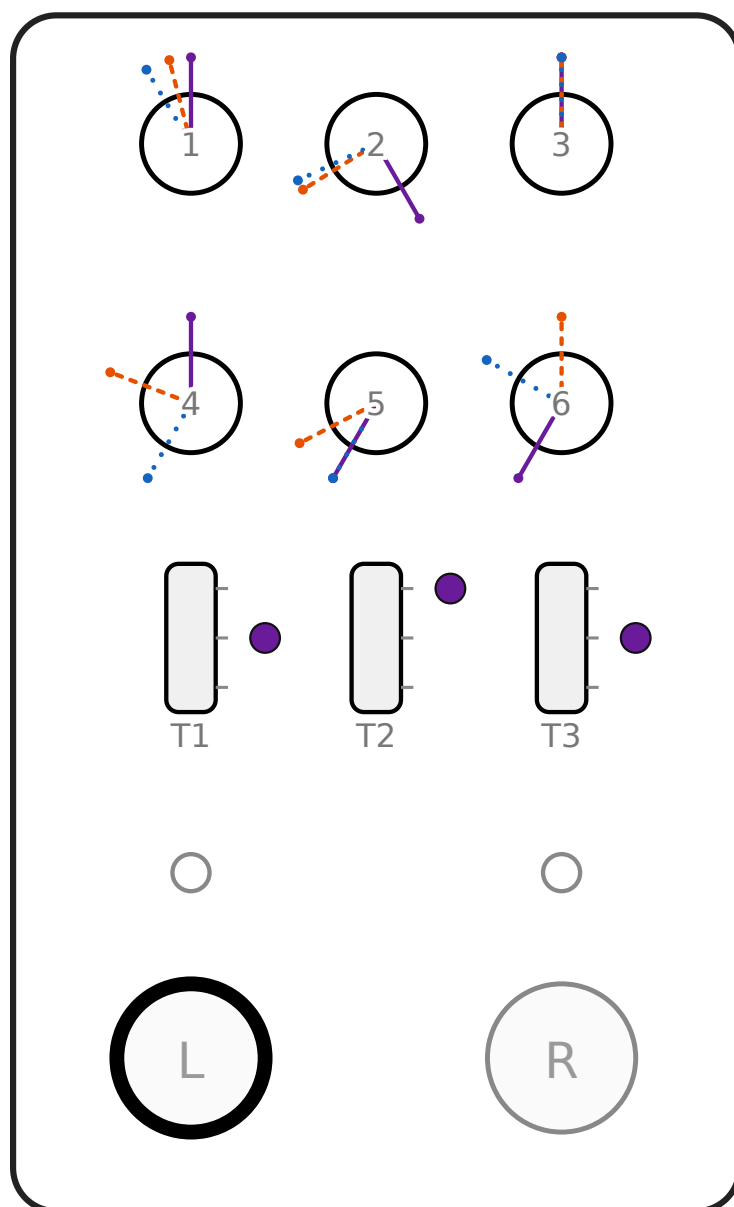
K1	F3	3111.7 Hz	54%	12:22
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	11 cents	18%	8:50

● Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1	Up
T3	Gate	medium	Middle

Prince *Clenched teeth*

Factory: Set 1, LEFT slot



Formants well below Wukong under a wider detune, so it lands darker and rougher. Reads as effort rather than power.

Prince: every position

● Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

● Menu 2 (hold RIGHT)

K1	F1	741.6 Hz	45%	11:31
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1066 Hz	27%	9:42
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

● Menu 3 (hold LEFT)

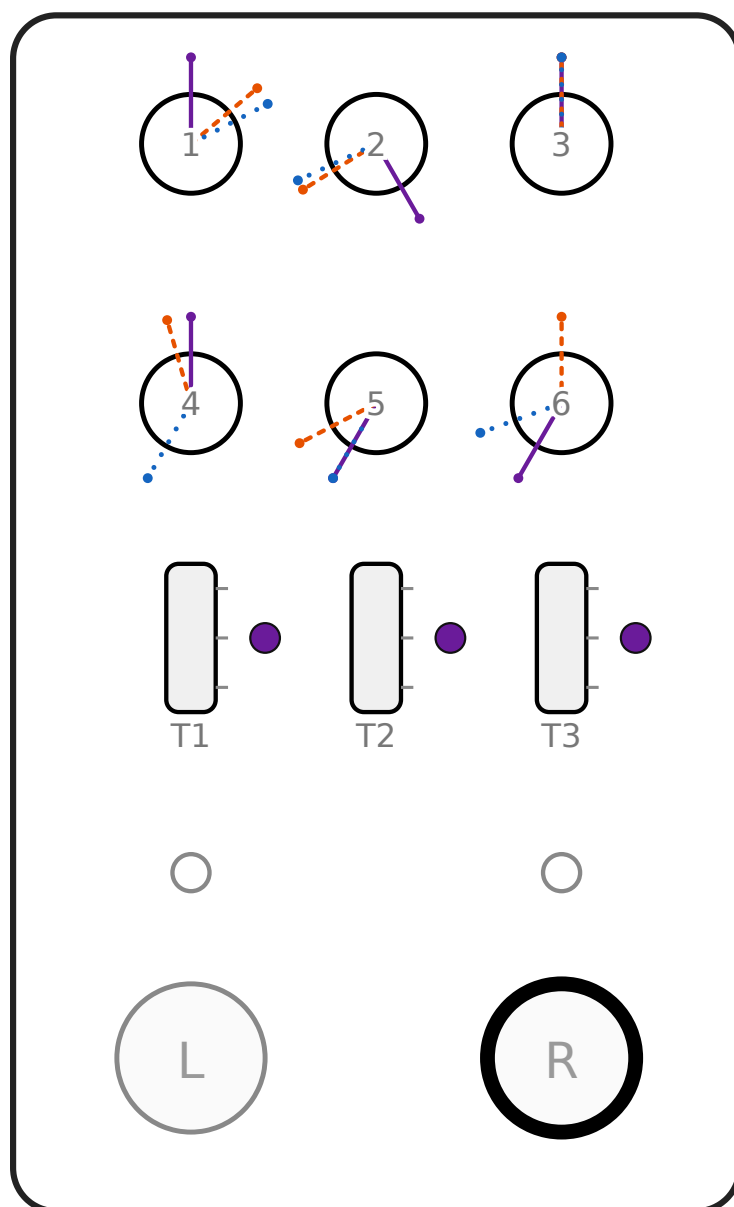
K1	F3	2688.3 Hz	40%	10:58
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	18 cents	30%	10:00

● Toggles

T1	Octave	0	Middle
T2	Memory page	Set 1	Up
T3	Gate	medium	Middle

Rice *Clean and glassy*

Factory: Set 2, RIGHT slot



The highest F3 of the eight over the narrowest detune in the factory set (tied with Ki-Ki), barely spread. Bright and hard-edged. Cuts without sounding strained.

Rice: every position

● Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

● Menu 2 (hold RIGHT)

K1	F1	1000 Hz	67%	1:40
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1437.5 Hz	45%	11:28
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

● Menu 3 (hold LEFT)

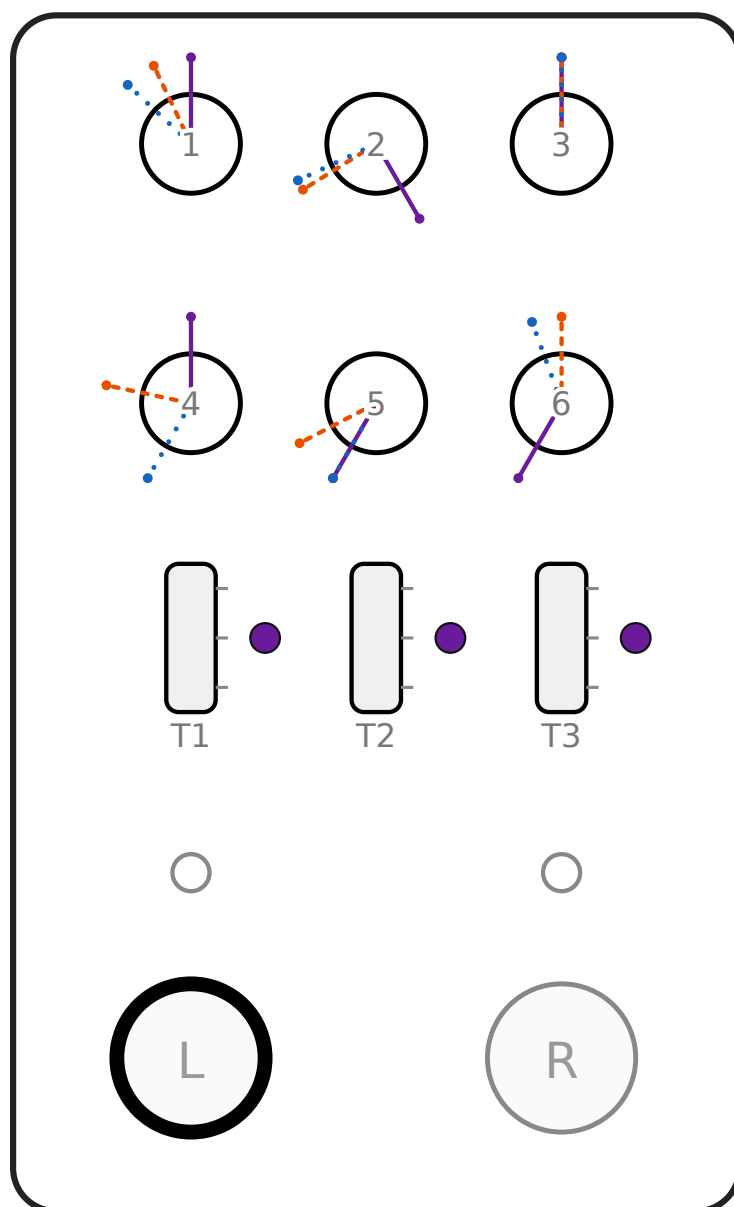
K1	F3	3625 Hz	71%	2:05
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	8 cents	13%	8:20

● Toggles

T1	Octave	0	Middle
T2	Memory page	Set 2	Middle
T3	Gate	medium	Middle

Flute *Low growl*

Factory: Set 2, LEFT slot



The lowest F2 of the eight, second only to Master on F1 and F3, under a wide detune. Thick and throaty rather than piercing.

Flute: every position

Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

Menu 2 (hold RIGHT)

K1	F1	697.6 Hz	41%	11:09
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1002.8 Hz	24%	9:24
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

Menu 3 (hold LEFT)

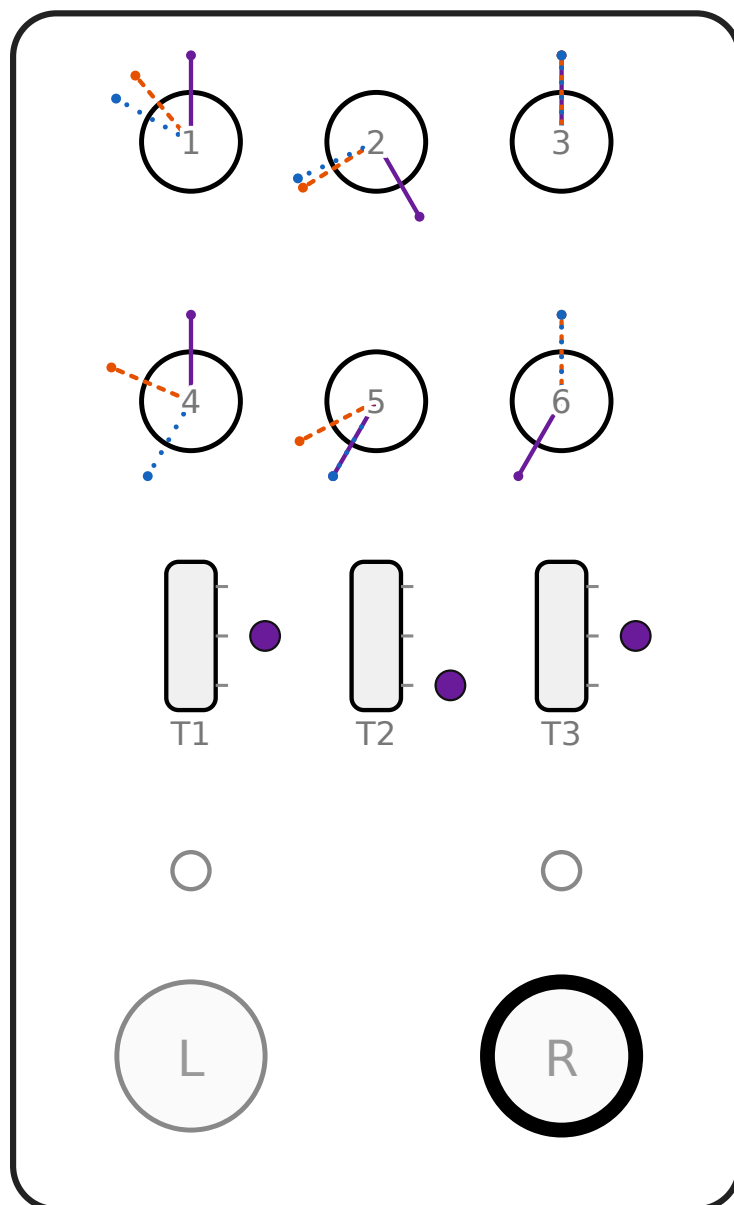
K1	F3	2528.8 Hz	34%	10:26
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	26 cents	43%	11:20

Toggles

T1	Octave	0	Middle
T2	Memory page	Set 2	Middle
T3	Gate	medium	Middle

Master *Old and graveled*

Factory: Set 3, RIGHT slot



The lowest F1 and F3 of the eight, the widest detune of any card. Dark, frayed, faintly ridiculous.

Master: every position

● Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

● Menu 2 (hold RIGHT)

K1	F1	640 Hz	37%	10:40
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1080 Hz	28%	9:46
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

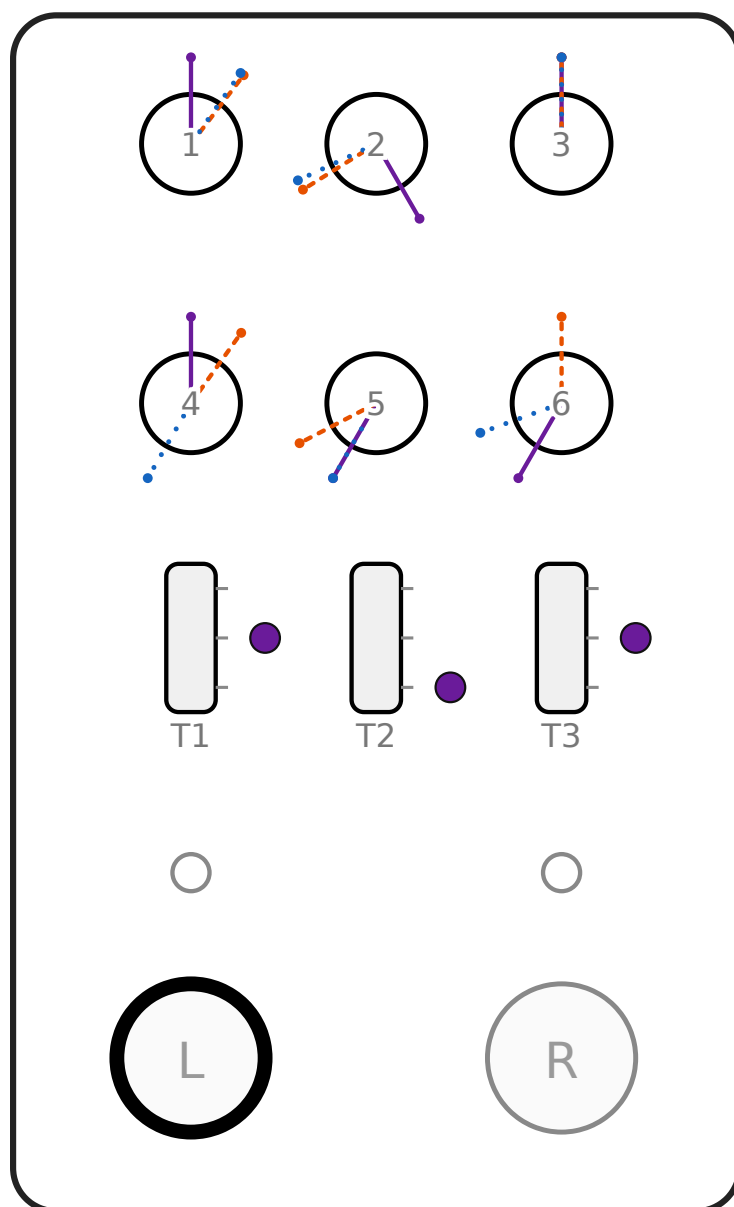
● Menu 3 (hold LEFT)

K1	F3	2400 Hz	30%	10:00
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	30 cents	50%	12:00

● Toggles

T1	Octave	0	Middle
T2	Memory page	Set 3	Down
T3	Gate	medium	Middle

Factory: Set 3, LEFT slot



The highest F2 of the eight and, with Rice, the narrowest detune, all three formants crowded high with nothing blunting them. Shrill, nasal, annoyed.

Ki-Ki: every position

● Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

● Menu 2 (hold RIGHT)

K1	F1	950 Hz	62%	1:15
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1800 Hz	62%	1:11
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

● Menu 3 (hold LEFT)

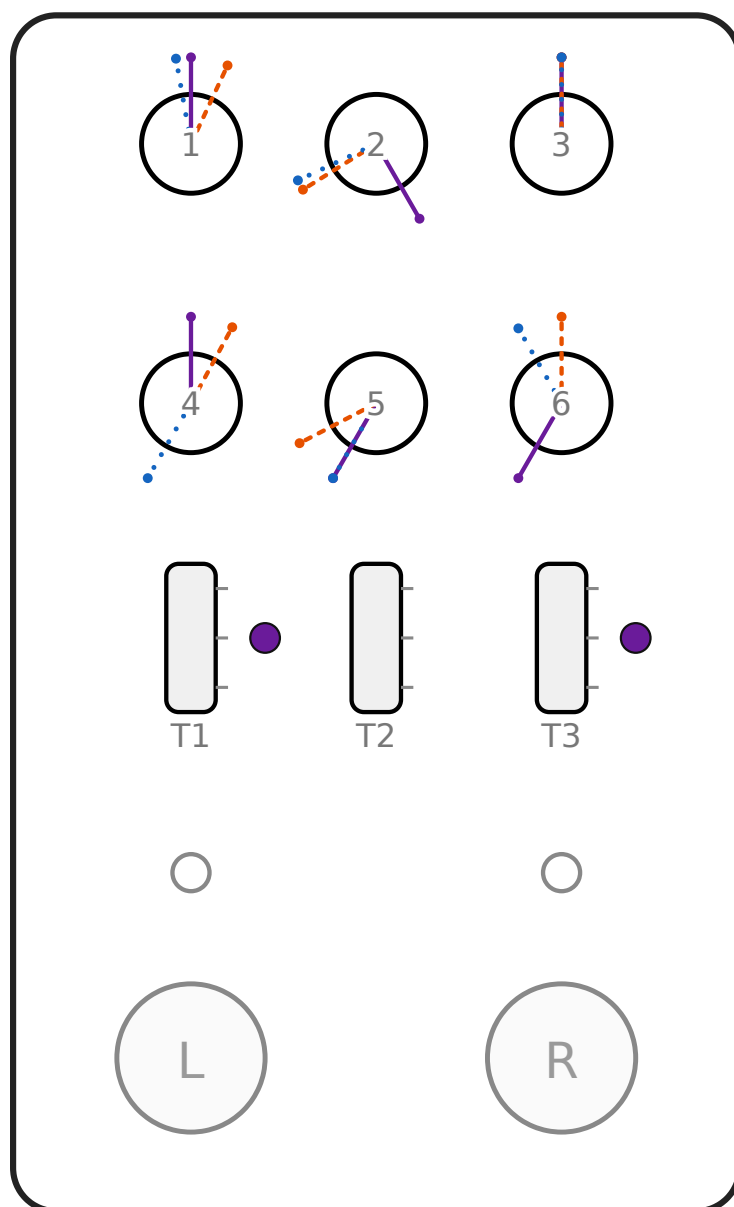
K1	F3	3350 Hz	62%	1:10
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	8 cents	13%	8:20

● Toggles

T1	Octave	0	Middle
T2	Memory page	Set 3	Down
T3	Gate	medium	Middle

Boo *Rubbery and hollow*

No factory slot: save it to any page



A wide F1-to-F2 gap with F3 close above F2, thickened by a wide detune. A big soft body rather than an edge.

Boo: every position

Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

Menu 2 (hold RIGHT)

K1	F1	900 Hz	58%	12:50
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1750 Hz	60%	12:57
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

Menu 3 (hold LEFT)

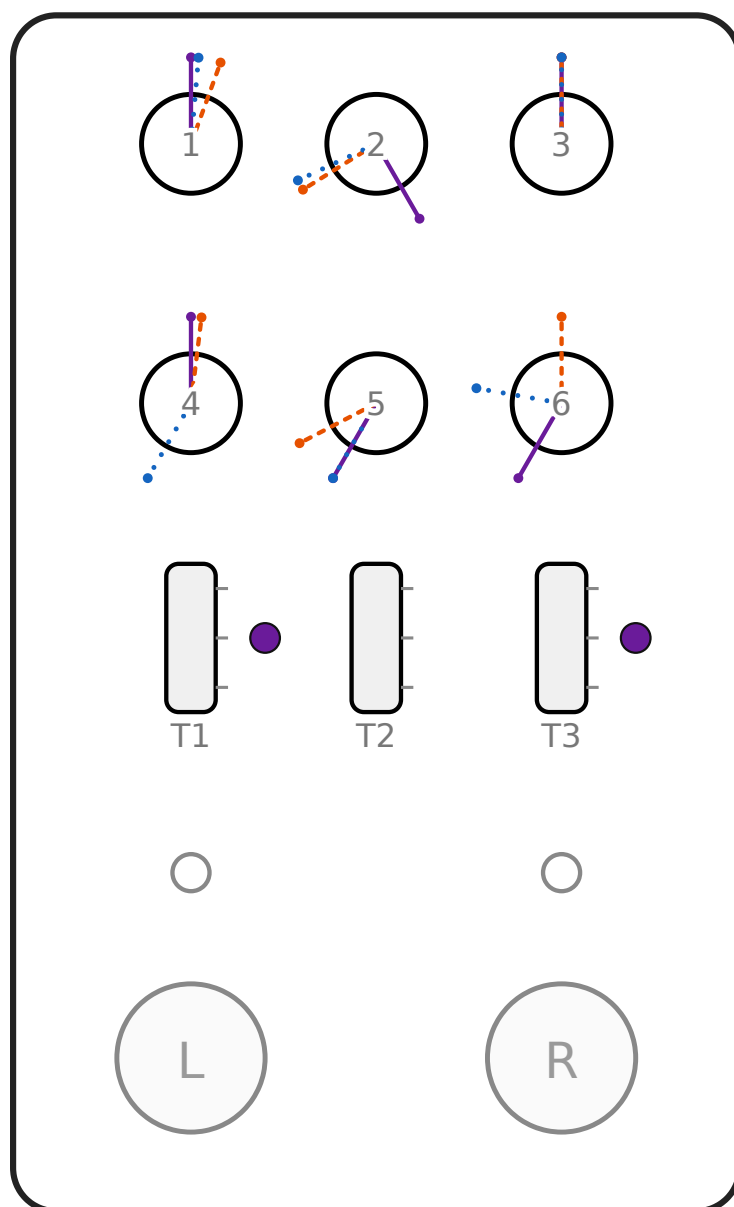
K1	F3	2900 Hz	47%	11:40
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	24 cents	40%	11:00

Toggles

T1	Octave	0	Middle
T2	Memory page	any Set	any
T3	Gate	medium	Middle

Fling *Light and quick*

No factory slot: save it to any page



F1 and F3 close to Wukong's with F2 lifted well above, under a modest detune so the edge stays on the formants. The most usable of the eight under a band.

Fling: every position

● Menu 1 (default)

K1	Vocal vol	unity	50%	12:00
K2	Mix	full voice	100%	5:00
K3	Master vol	unity	50%	12:00
K4	Tone	flat	50%	12:00
K5	Glide	0 ms	0%	7:00
K6	Vocal size	as written	0%	7:00

● Menu 2 (hold RIGHT)

K1	F1	880 Hz	57%	12:40
K2	F1 bandwidth	32.5 Hz	9%	7:56
K3	F1 amount	1.0	50%	12:00
K4	F2	1600 Hz	52%	12:14
K5	F2 bandwidth	47.5 Hz	11%	8:05
K6	F2 amount	1.0	50%	12:00

● Menu 3 (hold LEFT)

K1	F3	3050 Hz	52%	12:10
K2	F3 bandwidth	62.5 Hz	12%	8:10
K3	F3 amount	1.0	50%	12:00
K4	Vib	0 Hz	0%	7:00
K5	Depth	0 cents	0%	7:00
K6	Detune	14 cents	23%	9:20

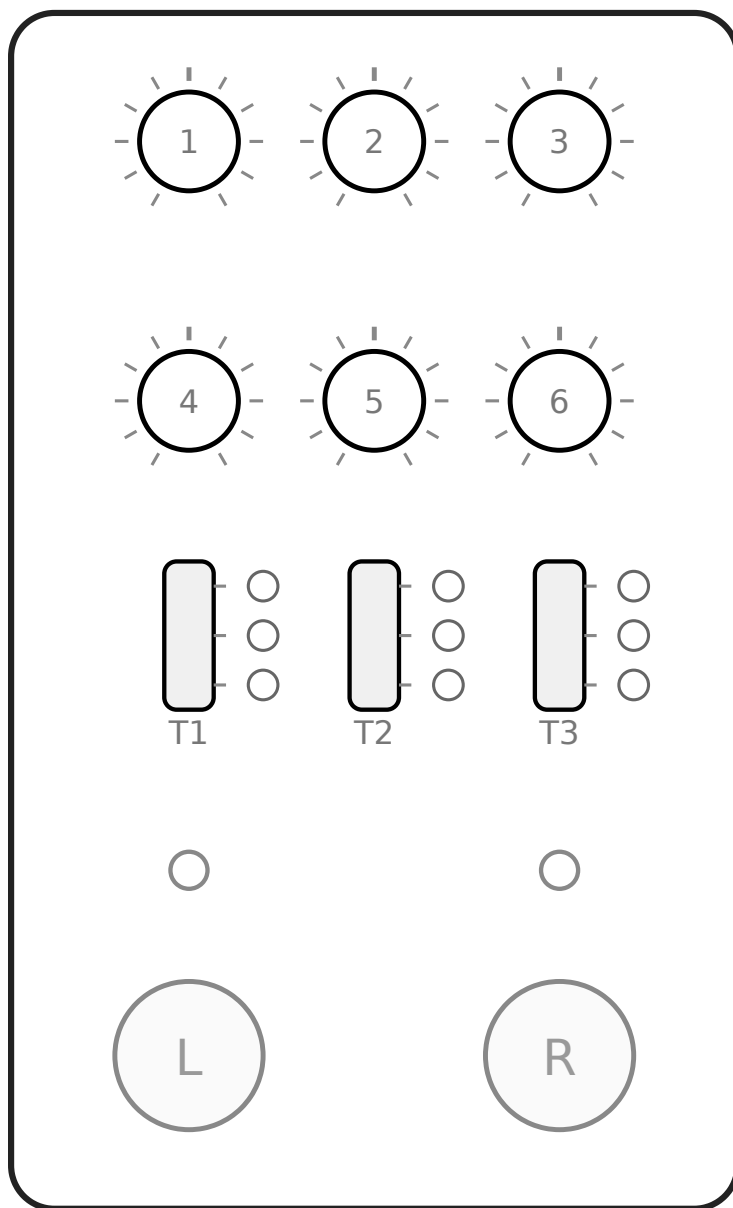
● Toggles

T1	Octave	0	Middle
T2	Memory page	any Set	any
T3	Gate	medium	Middle

Your own voice

Name _____

Saved to Set _____, _____ slot



Pointers: solid for menu 1, dashed 2, dotted 3. One dot per toggle.

Your own voice: every position

● Menu 1 (default)

	Value	Clock
K1 Vocal vol		
K2 Mix		
K3 Master vol		
K4 Tone		
K5 Glide		
K6 Vocal size		

● Menu 2 (hold RIGHT)

K1 F1		
K2 F1 bandwidth		
K3 F1 amount		
K4 F2		
K5 F2 bandwidth		
K6 F2 amount		

● Menu 3 (hold LEFT)

K1 F3		
K2 F3 bandwidth		
K3 F3 amount		
K4 Vib		
K5 Depth		
K6 Detune		

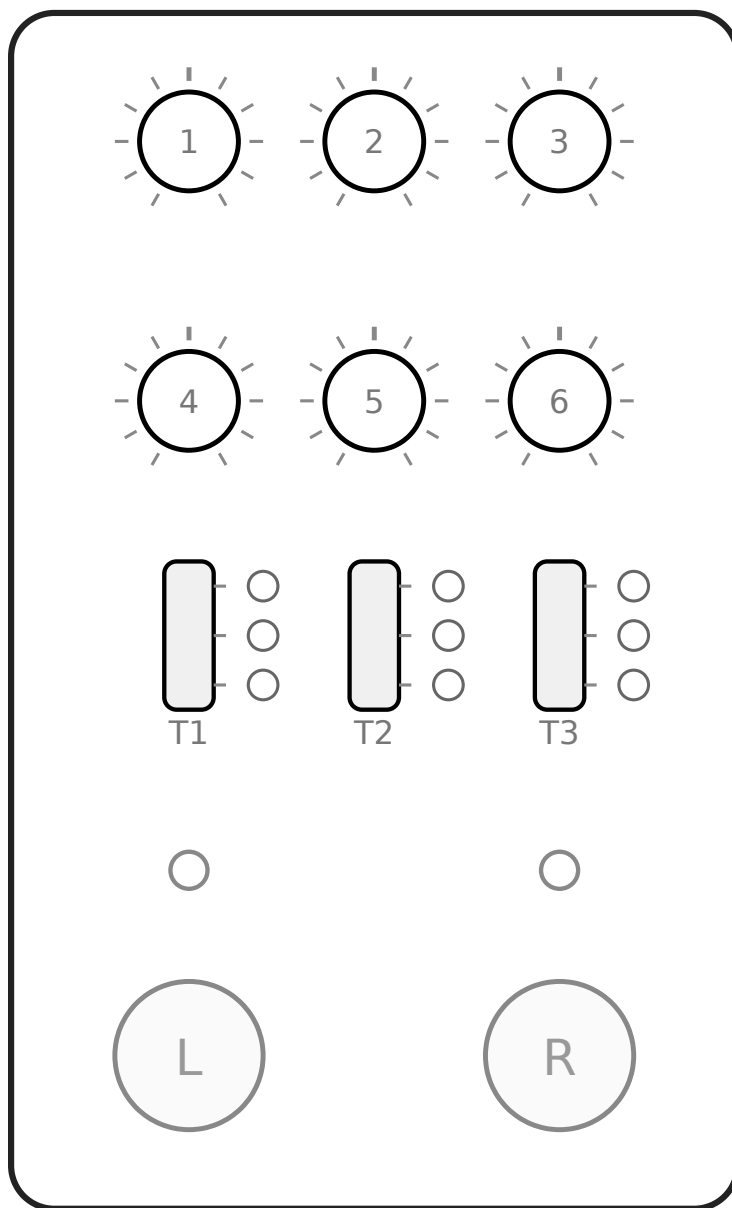
● Toggles

T1 Octave		
T2 Memory page		
T3 Gate		

Your own voice

Name _____

Saved to Set _____, _____ slot



Pointers: solid for menu 1, dashed 2, dotted 3. One dot per toggle.

Your own voice: every position

● Menu 1 (default)

	Value	Clock
K1 Vocal vol		
K2 Mix		
K3 Master vol		
K4 Tone		
K5 Glide		
K6 Vocal size		

● Menu 2 (hold RIGHT)

K1 F1		
K2 F1 bandwidth		
K3 F1 amount		
K4 F2		
K5 F2 bandwidth		
K6 F2 amount		

● Menu 3 (hold LEFT)

K1 F3		
K2 F3 bandwidth		
K3 F3 amount		
K4 Vib		
K5 Depth		
K6 Detune		

● Toggles

T1 Octave		
T2 Memory page		
T3 Gate		

12 Specifications

Platform	Cleveland Music Co. Hothouse, Daisy Seed3, 125B enclosure
Processing	48 kHz, 128-sample blocks (2.7 ms), 480 MHz
Voice	FOF [formant wave function] formant-grain synthesis, three formants
Tracking	Cycfi Q bitstream autocorrelation, monophonic
Bypass	Buffered (digital pass-through), 10 ms crossfade both ways
Power	9 V DC, 2.1 mm barrel, center negative
Memory	6 voice slots, the global charge configuration and the chord mode setting, kept across power cycles

Firmware update

From **bypass**, press both footswitches together and hold about 2 seconds. Press them at once: one held a second or more first also saves over that side's slot. The pedal appears over USB [universal serial bus] in DFU [Device Firmware Upgrade] mode. Flash it at withakerik.github.io/db-scream-z/flash.html in Chrome or Edge, or with dfu-util.

Credits

Formant voice synthesis derived from MonkSynth / Delay Lama. Pitch detection by cycfi/q, Boost Software License 1.0. Hardware from [clevelandmusicco/HothouseExamples](https://clevelandmusicco.com/HothouseExamples), open source hardware under CC BY-SA 4.0 [Creative Commons Attribution-ShareAlike 4.0]. Firmware: GPL-3.0 [General Public License, version 3]. The voices are synthesized from measured formant coefficients; no audio is redistributed.

Personal, non-commercial build. The names here are ours and refer to nothing.



Thanks for the great idea Tucker!